ALGEBRA

बीजगणित

SHEET - 01

CLASS ROOM SHEETS / FOR ALL EXAMS

UPDATED

ALGEBRA (बीजगणित)

CLASSROOM SHEET-01

1. If $x + \frac{1}{x} = 3$ Find

(i) $x^2 + \frac{1}{x^2}$ (ii) $x^3 + \frac{1}{x^3}$

(iii) $x^4 + \frac{1}{x^4}$ (iv) $x^5 + \frac{1}{x^5}$

(v) $x^6 + \frac{1}{x^6}$ (vi) $x^7 + \frac{1}{x^7}$

2. If $x - \frac{1}{x} = 4$, Find

(i) $x^2 + \frac{1}{x^2}$ (ii) $x^2 - \frac{1}{x^2}$

(iii) $x^3 - \frac{1}{x^3}$ (iv) $x^4 + \frac{1}{x^4}$

(v) $x^5 - \frac{1}{x^5}$ (vi) $x^6 + \frac{1}{x^6}$

3. If $x + \frac{1}{x} = 7$, find the value of $x^3 + \frac{1}{x^3}$.

SSC Phase-XII 25/06/2024 (Shift-04)

(a) 343 (b) 332
(c) 340 (d) 322

4. If $a + \frac{1}{a} = 3$, then the value of $a^4 + \frac{1}{a^4}$ is:

SSC CPO 04/10/2023 (Shift-01)

(a) 27 (b) 81
(c) 48 (d) 47

5. If $a + \frac{1}{a} = 7$, then $a^5 + \frac{1}{a^5}$ is equal to:

SSC CGL 17/07/2023 (Shift-02)

(a) 15127 (b) 13127
(c) 14527 (d) 11512

6. If $x + \frac{1}{x} = -6$, what will be the value of $x^5 + \frac{1}{x^5}$

SSC CGL 17/07/2023 (Shift-03)

(a) –7776 (b) –6726
(c) –6730 (d) –6732

7. If $\left(a + \frac{1}{a}\right) = 7\sqrt{3}$, then what is the value of $(a^6 + a^{-6})$?

SSC CGL 19/09/2024 (Shift-01)

(a) 3048190 (b) 3048542
(c) 3048132 (d) 3048625

8. If $\left(x - \frac{1}{x}\right) = 10$, what is the value of $x^4 + \frac{1}{x^4}$?

SSC CGL 26/07/2023 (Shift-03)

(a) 10404 (b) 10402
(c) 10406 (d) 10400

9. If $x - \frac{1}{x} = 10$, then $x^3 - \frac{1}{x^3}$ is equal to :

SSC CHSL 04/07/2019 (Shift-02)

(a) 970 (b) 1000
(c) 1030 (d) 1100

10. If $\left(3y - \frac{3}{y}\right) = 5$, find the value of $\left(y^2 + \frac{1}{y^2}\right)$.

SSC CGL 19/07/2023 (Shift-02)

(a) $\frac{47}{9}$ (b) $\frac{49}{9}$
(c) $\frac{41}{9}$ (d) $\frac{43}{9}$

11. If $x - \frac{1}{x} = 13$, what will be the value of $x^4 + \frac{1}{x^4}$?

SSC CGL 03/12/2022 (Shift-01)

(a) 28561 (b) 29243
(c) 27887 (d) 29239

12. If $\left(x - \frac{1}{x}\right) = 5$, then the value of $\left(x^5 - \frac{1}{x^5}\right)$ is :

SSC CHSL 07/08/2023 (Shift-02)

(a) 3775 (b) 3740
(c) 3715 (d) 3725

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13. If $x - \frac{1}{x} = -6$, what will be the value of $x^5 - \frac{1}{x^5}$?

SSC CGL 18/07/2023 (Shift-02)

(a) – 8898 (b) – 8896
(c) – 8886 (d) – 8892

14. If $\left(y - \frac{1}{y}\right) = 4$, find the value of $\left(y^6 + \frac{1}{y^6}\right)$.

SSC CGL 25/07/2023 (Shift-01)

(a) 5774 (b) 4096
(c) 5776 (d) 5778

15. If $x > 1$ and $x^2 + \frac{1}{x^2} = 83$ then, $x^3 - \frac{1}{x^3}$ is:

SSC CGL 17/09/2024 (Shift-01)

(a) 884 (b) 876
(c) 754 (d) 756

16. If $\sqrt{x} + \frac{1}{\sqrt{x}} = 7$, then the value of $x + \frac{1}{x}$ is equal to:

SSC CGL 10/09/2024 (Shift-02)

(a) 47 (b) 49
(c) 51 (d) 45

17. If $\sqrt{x} - \frac{1}{\sqrt{x}} = 4$, then $x^2 + \frac{1}{x^2}$ is equal to :

SSC CGL 10/06/2019 (Shift-01)

(a) 192 (b) 326
(c) 322 (d) 256

18. If $\sqrt{x} - \frac{1}{\sqrt{x}} = \sqrt{6}$, then $x^2 + \frac{1}{x^2}$ is equal to :

SSC CGL 11/06/2019 (Shift-02)

(a) 62 (b) 40
(c) 54 (d) 66

19. If $\left(x + \frac{1}{x}\right) = 7$, $x > 0$. The positive value of $\left(x - \frac{1}{x}\right)$ is:

SSC CGL 19/09/2024 (Shift-02)

(a) $5\sqrt{4}$ (b) $5\sqrt{5}$
(c) $3\sqrt{5}$ (d) $4\sqrt{5}$

20. If x is an integar such that $x + \frac{1}{x} = \frac{65}{8}$, then find the value of $x - \frac{1}{x}$.

SSC CGL 17/09/2024 (Shift-03)

(a) $\frac{63}{6}$ (b) $\frac{63}{8}$
(c) $\frac{36}{8}$ (d) $\frac{32}{6}$

21. If $\left(x - \frac{1}{x}\right) = 0.4$, and x>0, what is the value of $\left(x^2 - \frac{1}{x^2}\right)$?

SSC CHSL 07/08/2023 (Shift-02)

(a) $\frac{4}{25}\sqrt{26}$ (b) $\frac{2}{25}\sqrt{29}$
(c) $\frac{7}{25}\sqrt{26}$ (d) $\frac{3}{25}\sqrt{29}$

22. If $x + \frac{1}{x} = 2\sqrt{10}$ where $x > 1$ then the value of $\left(x^3 - \frac{1}{x^3}\right)$ is:

SSC CGL 11/09/2024 (Shift-01)

(a) 221 (b) 234
(c) 198 (d) 216

23. If $\left(x^2 + \frac{1}{x^2}\right) = 6$ and $0 < x < 1$, what is the value of $x^4 - \frac{1}{x^4}$?

SSC CGL 26/07/2023 (Shift-01)

(a) $24\sqrt{2}$ (b) $-24\sqrt{2}$
(c) $-12\sqrt{10}$ (d) $12\sqrt{10}$

24. If $x^2 - \frac{1}{x^2} = 4\sqrt{2}$, what is the value of $x^4 - \frac{1}{x^4}$?

SSC CGL 27/07/2023 (Shift-02)

(a) $16\sqrt{2}$ (b) $8\sqrt{2}$
(c) $24\sqrt{2}$ (d) $32\sqrt{2}$

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25. If $\left(x+\frac{1}{x}\right)=5\sqrt{2}$, and $x>1$, what is the value of $\left(x^6-\frac{1}{x^6}\right)$?

SSC CHSL 02/08/2023 (Shift-02)

(a) $22970\sqrt{23}$ (b) $23030\sqrt{23}$

(c) $23060\sqrt{23}$ (d) $22960\sqrt{23}$

26. If $\left(x-\frac{1}{x}\right)=2\sqrt{2}$, and $x>1$, what is the value of $\left(x^6-\frac{1}{x^6}\right)$.

SSC CGL 20/07/2023 (Shift-03)

(a) $372\sqrt{6}$ (b) $384\sqrt{6}$

(c) $396\sqrt{6}$ (d) $420\sqrt{6}$

27. If $\left(x+\frac{1}{x}\right)=5$, and $x>1$, what is the value of $\left(x^8-\frac{1}{x^8}\right)$?

SSC CGL 18/07/2023 (Shift-03)

(a) $60605\sqrt{21}$ (b) $60615\sqrt{21}$

(c) $60705\sqrt{21}$ (d) $60725\sqrt{21}$

28. If $x+\frac{1}{x}=15$, then the value of $\frac{7x^2-9x+7}{x^2-x+1}$ is:

SSC CGL 12/09/2024 (Shift-01)

(a) $-\frac{48}{7}$ (b) $-\frac{22}{7}$

(c) $\frac{48}{7}$ (d) $\frac{22}{7}$

29. If $x+\frac{1}{x}=10$, then find the value of $\frac{7x}{x^2+1-8x}$.

SSC CHSL 03/06/2022 (Shift- 01)

(a) 3.5 (b) 4.5

(c) 2.5 (d) 5.5

30. If $x+\frac{1}{x}=5$, $x\neq 0$ then the value of $\frac{x^4+\frac{1}{x^2}}{x^2-3x+1}$ is equal to :

SSC CHSL 20/10/2020 (Shift- 01)

(a) 55 (b) 60

(c) 65 (d) 50

31. If $x-\frac{1}{x}=5$, $x\neq 0$, then what is the value of $\frac{x^6+3x^3-1}{x^6-8x^3-1}$?

SSC CGL TIER II 16/11/2020

(a) $\frac{13}{12}$ (b) $\frac{11}{13}$

(c) $\frac{3}{8}$ (d) $\frac{4}{9}$

32. If $x^2+3x+1=0$, then what is the value of $x^6+\frac{1}{x^6}$?

SSC CGL 3/03/2020 (Shift- 02)

(a) 324 (b) 322

(c) 318 (d) 327

33. If $x^2-2\sqrt{5}x+1=0$, then what is the value of $x^5+\frac{1}{x^5}$?

SSC CGL 5/03/2020 (Shift- 01)

(a) $610\sqrt{5}$ (b) $408\sqrt{5}$

(c) $612\sqrt{5}$ (d) $406\sqrt{5}$

34. If $\frac{x}{x^2-2x+1}=\frac{1}{3}$, then the value of $x^3+\frac{1}{x^3}$ is:

SSC CGL 26/09/2024 (Shift-03)

(a) 110 (b) 130

(c) 160 (d) 180

35. If $x^2-5x+1=0$, then the value of $\frac{x^6+x^4+x^2+1}{5x^3}=?$

SSC CGL 01/12/2022 (Shift-01)

(a) 30 (b) 25

(c) 23 (d) 28

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36. If $x^2 - 3x + 1 = 0$, then the value of $\frac{x^4 + \frac{1}{x^2}}{x^2 + 5x + 1}$ is:

SSC CGL MAINS 03/02/2022

(a) $\frac{9}{4}$ (b) $\frac{27}{8}$

(c) $\frac{5}{2}$ (d) 2

37. If for a non-zero x, $7x^2 + 5x + 7 = 0$, then the $x^3 + \frac{1}{x^3}$ is ____.

SSC Phase-XII 24/06/2024 (Shift-02)

(a) $\frac{420}{343}$ (b) $\frac{610}{420}$

(c) $\frac{610}{343}$ (d) $\frac{343}{610}$

38. If $x^2 + 7x + 8 = 0$, then find the value of $\frac{4x}{x^2 - 5x + 8}$

SSC CGL 25/09/2024 (Shift-01)

(a) $\frac{1}{2}$ (b) $\frac{-1}{4}$

(c) $\frac{-1}{3}$ (d) $\frac{1}{6}$

39. If $x^2 - 3x - 1 = 0$, then the value of $(x^2 + 8x - 1)(x^3 + x^{-1})^{-1}$ is :

SSC CGL 07/06/2019 (Shift- 02)

(a) $\frac{3}{8}$ (b) 8

(c) 1 (d) 3

40. If $x + \frac{1}{9x} = 3$, then the value of $9x^2 + \frac{1}{9x^2}$ is:

SSC CGL 11/09/2024 (Shift-03)

(a) 87 (b) 81

(c) 77 (d) 79

41. If $5x + \frac{1}{3x} = 5$, find the value of $9x^2 + \frac{1}{25x^2}$

(a) 0 (b) 5

(c) 4 (d) 7.8

42. If $3x + \frac{1}{2x} = 5$ then find $8x^3 + \frac{1}{27x^3}$

(a) $118\frac{1}{2}$ (b) $30\frac{10}{27}$

(c) 0 (d) 1

43. If $x + \frac{1}{16x} = 3$, then the value of $16x^3 + \frac{1}{256x^3}$ is :

SSC CGL TIER II 12/09/2019

(a) 423 (b) 441

(c) 432 (d) 414

44. If $5x + \frac{1}{3x} = 4$, then the value of $9x^2 + \frac{1}{25x^2}$ is :

SSC CGL 04/03/2020 (Shift- 01)

(a) $\frac{174}{125}$ (b) $\frac{144}{125}$

(c) $\frac{114}{25}$ (d) $\frac{119}{25}$

45. If $20x^2 - 30x + 1 = 0$, then what is the value of $25x^2 + \frac{1}{16x^2}$ is :

SSC CGL 05/03/2020 (Shift-02)

(a) $53\frac{1}{2}$ (b) $58\frac{1}{2}$

(c) $53\frac{3}{4}$ (d) $58\frac{3}{4}$

46. If $4a + \frac{1}{5a} = 4$, then the value of $25a^2 + \frac{1}{16a^2}$ is :

SSC CHSL 21/10/2020 (Shift-03)

(a) $\frac{45}{2}$ (b) $\frac{55}{2}$

(c) $\frac{43}{2}$ (d) $\frac{45}{4}$

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47. If $2x^2 + 5x + 1 = 0$, then one of the values of $x - \frac{1}{2x}$?

SSC CGL MAINS 29/01/2022

(a) $\frac{\sqrt{17}}{2}$ (b) $\frac{13}{2}$

(c) $\frac{5}{2}$ (d) $\frac{\sqrt{13}}{2}$

48. If $5x - \frac{1}{4x} = 6$, $x > 0$, then find the value of $25x^2 - \frac{1}{16x^2}$.

SSC CGL 21/04/2022 (Shift 01)

(a) $6\sqrt{41}$ (b) 36

(c) $\sqrt{246}$ (d) $6\sqrt{31}$

49. If $\left(4x^2 + \frac{1}{4x^2}\right) = 2$,than what is the value of $\left(8x^3 + \frac{1}{8x^3}\right)$?

(a) +2 (b) –2

(c) +4 (d) –4

50. If $\left(0.4x + \frac{1}{x}\right) = 5$, what is the value of $\left(0.064x^3 + \frac{1}{x^3}\right)$?

SSC CGL 20/04/2022 (Shift 01)

(a) 119 (b) 125

(c) 110 (d) 105

51. If $16x^4 + \frac{1}{16x^4} = 14159$, then find the value of $2x + \frac{1}{2x}$.

SSC CPO 27/06/2024 (Shift-03)

(a) 9 (b) 11

(c) 12 (d) 15

52. If $x^2 + \frac{1}{x^2} = 27$, then find

(i) $x + \frac{1}{x}$ (ii) $x - \frac{1}{x}$

53. If $x^4 + \frac{1}{x^4} = 23$, find

(i) $x + \frac{1}{x}$ (ii) $x^3 + \frac{1}{x^3}$

54. If $x^8 - 1442x^4 + 1 = 0$, then a possible value of $x - \frac{1}{x}$ is :

SSC CGL TIER-II (11/09/2019)

(a) 5 (b) 8

(c) 4 (d) 6

55. If $x > 0$ and $x^4 + \frac{1}{x^4} = 142$, what is the value of $x^7 + \frac{1}{x^7}$?

SSC CGL 20/07/2023 (Shift-02)

(a) $1561\sqrt{14}$ (b) $1563\sqrt{14}$

(c) $1560\sqrt{14}$ (d) $1562\sqrt{14}$

56. If $x > 0$, and $x^4 + \frac{1}{x^4} = 2207$, what is the value of $x^7 + \frac{1}{x^7}$?

SSC CGL 02/12/2022 (Shift-01)

(a) 710649 (b) 710647

(c) 710654 (d) 710661

57. If $x^4 + x^{-4} = 47$, $(x > 0)$, then the value of $(2x - 3)^2$ is :

SSC CHSL 8/07/2019 (Shift-03)

(a) 2 (b) 3

(c) 5 (d) 4

58. If $x^4 + x^{-4} = 194$, $(x > 0)$, then the value of $(2x - 4)^2$ is :

SSC CHSL 09/07/2019 (Shift-01)

(a) 15 (b) 20

(c) 12 (d) 16

59. If $x^4 + \frac{1}{x^4} = 322$, find $x^3 - \frac{1}{x^3}$

(a) ±76 (b) 76

(c) –76 (d) 95

60. If $x + \frac{1}{x} = 2$, find

(i) $x^{20} + x^{19} + x^{18} + \ldots\ldots + x^2 + x + 1$

(ii) $x^{49} - x^{48} + x^{47} - x^{46} + \ldots\ldots + x^3 - x^2 + x + 1$

(iii) $x^{12} + \frac{1}{x^{12}}$ (iv) $x^{17} + \frac{1}{x^9}$

(v) $x^{15} - \frac{1}{x^{17}}$

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61. If $x + \frac{1}{x} = 2$, then find the value of $\left(x^4 + \frac{4}{x^4}\right)\left(x^3 - \frac{4}{x^3}\right)$.

SSC CPO 27/06/2024 (Shift-03)

(a) –15 (b) 15
(c) –10 (d) 10

62. If $x + \frac{1}{x} = 2$, then find the value of $\left(x^{2723} + \frac{1}{x^{3356}}\right)$.

SSC CGL 18/09/2024 (Shift-02)

(a) 1 (b) 2
(c) 0 (d) –1

63. If $x + \frac{1}{x} = -2$

(i) $x^{39} + x^{38} + \ldots\ldots\ldots + x^2 + x + 1$

(ii) $x^{100} - x^{99} + x^{98} - x^{97} + \ldots\ldots + x^2 - x + 1$

(iii) $x^5 - \frac{1}{x^5}$

(iv) $x^{18} + \frac{1}{x^{18}}$ (v) $x^7 - \frac{1}{x^9}$

64. If $x + \frac{1}{x} = 1$, find

(i) $x^{18} + x^{12} + x^6 + 1$

(ii) $x^{99} + x^{96} + x^{93} + x^{90} + \ldots\ldots + x^6 + x^3 + 1$

(iii) $x^{45} - x^{42} + x^{39} - x^{36} + \ldots\ldots + x^9 - x^6 + x^3 + 1$

(iv) $x^{32} + x^{46} + x^{90} + 2$

65. If $x + \frac{1}{x} = \sqrt{3}$, find

(i) $x^{90} + \frac{1}{x^{90}}$ (ii) $x^{92} + \frac{1}{x^{92}}$

(iii) $x^{102} + x^{96} + x^{101} + x^{95} + x^{100} + x^{94}$

(iv) $x^6 - \frac{1}{x^6} + 2$ (v) $x^{18} + x^{12} + x^6 + 1$

66. If $\left(x + \frac{1}{x}\right)^2 = 3$, then what is the value of $x^6 + x^{-6}$?

SSC CGL MAINS (08/08/2022)

(a) 6 (b) 2
(c) –2 (d) –6

67. If $x + \frac{1}{x} = \sqrt{3}$, then the value of $x^{18} + x^{12} + x^6 + 1$ is :

SSC CHSL 21/10/2020 (Shift- 03)

(a) 0 (b) 2
(c) 3 (d) 1

68. If $x + \frac{1}{x} = -\sqrt{3}$ then $x^{67} + x^{53} + x^{43} + x^{29} + x^{24} + x^{12} + x^6 + 3$ is

(a) $\sqrt{3}$ (b) 0
(c) $2\left(2+\sqrt{3}\right)$ (d) $2\left(2-\sqrt{3}\right)$

69. If $x = 2 + \sqrt{3}$, find $x + \frac{1}{x}$

(a) 5 (b) 4
(c) –4 (d) $2\sqrt{3}$

70. If $x = 7 + 4\sqrt{3}$ then $x - \frac{1}{x}$

(a) $8\sqrt{3}$ (b) 4
(c) –4 (d) $-2\sqrt{3}$

71. If $x = 7 - 4\sqrt{3}$, then find the value of $\left(x^2 + \frac{1}{x^2}\right)$.

SSC CGL 18/09/2024 (Shift-01)

(a) 188 (b) 194
(c) 186 (d) 196

72. If $x = 3 + 2\sqrt{2}$, then the value of $\sqrt{x} - \frac{1}{\sqrt{x}}$ is:

SSC CHSL 26/10/2020 (Shift-03)

(a) 2 (b) 1
(c) 0 (d) 3

73. If $x = 1 + \sqrt{2}$, then the value of $\sqrt{x} + \left(\frac{1}{\sqrt{x}}\right)$ is:

SSC CHSL 17/03/2020 (Shift-03)

(a) 2.1014 (b) 2.1973
(c) 1.9876 (d) 1.9996

74. If $p = 7 + 4\sqrt{3}$, then what is the value of $\frac{p^6 + p^4 + p^2 + 1}{p^3}$?

SSC CGL 06/12/2022 (Shift-02)

(a) 2617 (b) 2167
(c) 2716 (d) 2176

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75. If $P = 7 + 4\sqrt{3}$ and $PQ = 1$, then what is the value of $\left(\frac{1}{P^2} + \frac{1}{Q^2}\right) = ?$

(a) 148 (b) 189
(c) 194 (d) 204

76. If $x = \frac{\sqrt{5}-\sqrt{4}}{\sqrt{5}+\sqrt{4}}$ and $y = \frac{\sqrt{5}+\sqrt{4}}{\sqrt{5}-\sqrt{4}}$ then the value of $\frac{x^2-xy+y^2}{x^2+xy+y^2} = ?$

SSC CGL 08/12/2022 (Shift-02)

(a) $\frac{361}{363}$ (b) $\frac{341}{343}$
(c) $\frac{384}{387}$ (d) $\frac{321}{323}$

77. If $m + \frac{1}{m-3} = 5$, then $(m-3)^8 + \frac{1}{(m-3)^8} = ?$

(a) 2 (b) 0
(c) –2 (d) 15

78. If $x + \frac{1}{x-9} = 11$, then $(x-11)^{12} + \frac{1}{(x-11)^{13}} = ?$

(a) 0 (b) 2
(c) –2 (d) 1

79. If $a + \frac{1}{a-4} = 6$, then $(a-3)^7 + \frac{1}{(a-7)^3} = ?$

(a) $63\frac{7}{8}$ (b) $255\frac{1}{8}$
(c) $127\frac{7}{8}$ (d) 216

80. If $x^2 - 22x + 111 = 0$, then what is the value of $(x-8)^2 - \frac{1}{(x-8)^2} = ?$

(a) $12\sqrt{10}$ (b) $8\sqrt{5}$
(c) $8\sqrt{3}$ (d) 18

81. If $x^2 + 13x + 39 = 0$, then $(x+8)^5 - \frac{1}{(x+8)^5} = ?$

(a) 393 (b) 396
(c) 392 (d) 394

82. If $x^2 - 12x + 33 = 0$, then what is the value of $(x-4)^4 + \frac{1}{(x-4)^4} = ?$

(a) 227 (b) 326
(c) 167 (d) 194

83. If $(x-a)(x-b) = 1$ and $a - b + 5 = 0$, then $(x-a)^3 - \frac{1}{(x-a)^3} = ?$

(a) 110 (b) 1
(c) 125 (d) 140

84. If $x^5 - \frac{1}{x^5} = \sqrt{7}$, then find $x^{35} - \frac{1}{x^{35}} = ?$

(a) $791\sqrt{7}$ (b) $789\sqrt{7}$
(c) $790\sqrt{7}$ (d) $792\sqrt{7}$

85. If $3\sqrt{x} + \frac{1}{2\sqrt{x}} = 3$, $x > 0$ then find $x^2(18x^2 - 7)$?

(a) $-\frac{1}{36}$ (b) $-\frac{2}{63}$
(c) $-\frac{1}{72}$ (d) $-\frac{8}{81}$

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ANSWER KEY

1.(i)7 (ii)18 (iii)47 (iv)123 (v)322 (vi)843									
2.(i)18 (ii) $8\sqrt{5}$ (iii)76 (iv)322 (v)1364 (vi)5778									
3.(d)	4.(d)	5.(a)	6.(b)	7.(a)	8.(b)	9.(c)	10.(d)	11.(d)	12.(a)
13.(c)	14.(d)	15.(d)	16.(a)	17.(c)	18.(a)	19.(c)	20.(b)	21.(a)	22.(b)
23.(a)	24.(c)	25.(b)	26.(c)	27.(a)	28.(c)	29.(a)	30.(a)	31.(a)	32.(b)
33.(a)	34.(a)	35.(c)	36.(a)	37.(c)	38.(c)	39.(c)	40.(d)	41.(d)	42.(b)
43.(a)	44.(c)	45.(c)	46.(a)	47.(a)	48.(a)	49.(d)	50.(a)	51.(b)	
52.(i) $\sqrt{29}$ (ii) 5		53.(i) $\sqrt{7}$	(ii) $4\sqrt{7}$	54.(d)	55.(a)	56.(b)	57.(c)	58.(c)	59.(a)
60.(i) 21	(ii) 2	(iii) 2	(iv) 2	(v) 0	61.(a)	62.(b)			
63.(i) 0	(ii)101	(iii) 0	(iv) 2	(v) 0					
64.(i) 4	(ii) 0	(iii) –14	(iv) 2						
65.(i) –2	(ii) –1	(iii) 0	(iv) 2	(v) 0	66.(c)	67.(a)	68.(c)	69.(b)	70.(a)
71.(b)	72.(a)	73.(b)	74.(c)	75.(c)	76.(d)	77.(a)	78.(a)	79.(c)	80.(a)
81.(a)	82.(d)	83.(d)	84.(a)	85.(c)					

Maths by Aditya Ranjan sir

CALCULATION TECHNIQUES

TECHNIQUE – 01



♦ Square Root of perfect square ♦

Calculate the square root of 5184

Step 01: The unit digit of 5184 is 4, so its square root will be end on 2 or 8.

Step 02: Now leave two digits from end and check the remaining number. In this, case it is 51.

$7^2 < 51 < 8^2$

So, the tens digit of square root is 7.

Step 03: Now the square root of 5184 can be 72 or 78. For this, we have to check whether 51 is closer to 7^2 or 8^2 As 51 is closer to 7^2.

Square root of 5184 is 72.



♦ Do it Yourself ♦

(i) $\sqrt{7744}$ = ...

(ii) $\sqrt{1764}$ = ...

(iii) $\sqrt{6561}$ = ...

(iv) $\sqrt{7396}$ = ...

(v) $\sqrt{9216}$ = ...

(vi) $\sqrt{4624}$ = ...

(vii) $\sqrt{7569}$ = ...

TECHNIQUE – 02



♦ Square Root of Non-perfect square ♦

$$\sqrt{N} = \sqrt{(A \pm B)} = \sqrt{A} \pm \frac{B}{2\sqrt{A}}$$

(First find a number which is perfect square near to the number N)

Ex: $\sqrt{10} = \sqrt{9+1} = \sqrt{9} + \frac{1}{2\sqrt{9}} = 3 + \frac{1}{2\times 3} = \frac{19}{6} = 3.17$

Ex: $\sqrt{14} = \sqrt{16-2} = \sqrt{16} - \frac{2}{2\sqrt{16}} = 4 - \frac{2}{2\times 4} = \frac{30}{8} = 3.75$



♦ Do it Yourself ♦

(i) $\sqrt{18}$ = ...

(ii) $\sqrt{29}$ = ...

(iii) $\sqrt{57}$ = ...

(iv) $\sqrt{265}$ = ...

(v) $\sqrt{567}$ = ...

SELF MENTORSHIP TABLE

	STEP-01	STEP-02	STEP-03	STEP-04	STEP-05	STEP-06
TOPIC NAME	VIDEOS	CLASSROOM SHEET	SSC MATHS	NOTES / FORMULA BOOK REVISION	CHAPTER TEST	PRO LEVEL PRACTICE SHEET
ALGEBRA SHEET-01						
Inverse Function (Basic Questions)						
Inverse Function (Quadratic Equation)						
Inverse Function (Unequal Co-efficient)						
Inverse Function (Reverse Case)						
Inverse Function (Special Cases)						
Inverse Function (Miscellaneous Questions)						